

TrES-1b

R-0213

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_180804 · created 2026-07-17T00:23:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458334.8924 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.021
Transit depth (Rp/Rs)^2	2.7 +/- 0.7 [%]
Orbital Inclination (inc)	85.54 +/- 0.86
Airmass coefficient 1 (a1)	1.369 +/- 0.018
Airmass coefficient 2 (a2)	-0.1712 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	None
Optimal Aperture	4.43
Optimal Annulus	6.44
Transit Duration (day)	0.073 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.021 vs 0.1358 (deviation 0.92σ)
Duration fit vs published	0.073 d vs 0.1042 d (deviation 1.25σ)
Mid-time O–C	-5.1 min
Depth SNR	5.3
Residual scatter	1.27 %

Light curve

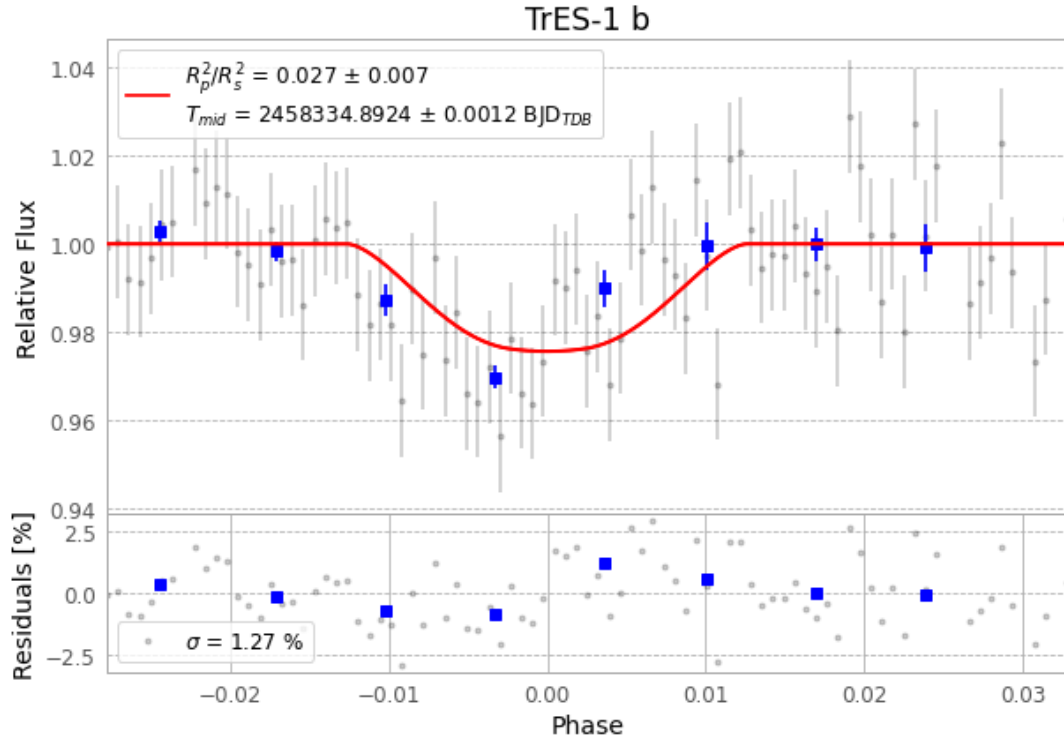


Figure 1 — FinalLightCurve_TrES-1 b_2018-08-04.png (SHA-256 307f7411ebf430e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7ca50924da1cbe5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

87 FITS frames (57 MB), TRES-1180804071812.FITS → TRES-1180804114212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-180804.log (763be83ce345...), FinalLightCurve_TrES-1 b_2018-08-04.png (307f7411ebf4...), FinalLightCurve_TrES-1 b_2018-08-04.csv (67519556fbcb...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:23Z.